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REFRIGERATION CYCLE SYSTEM

Abstract

Power consumption is reduced in a refrigeration cycle system including a cascade heat exchanger that causes heat exchange between a CO.sub.2 refrigerant and a refrigerant other than CO.sub.2, the refrigeration cycle system implementing a first refrigeration cycle using the CO.sub.2 refrigerant and a second refrigeration cycle using the refrigerant other than CO.sub.2. A controller controls a first compressor and a second compressor such that an intermediate temperature T_m between an evaporation temperature of a first refrigerant and a condensation temperature of a second refrigerant in a cascade heat exchanger satisfies $T_2 + (T_1 - T_2) \times 0.1 \leq T_m \leq T_2 + (T_1 - T_2) \times 0.4$, where T_1 ° C. represents a pressure equivalent saturation temperature of a refrigerant discharged from the first compressor, and T_2 ° C. represents a pressure equivalent saturation temperature of a refrigerant sucked into the second compressor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a bypass continuation of International Application No. PCT/JP2023/042224, filed Nov. 24, 2023, which claims priority to Japanese patent application JP 2022-188574, filed Nov. 25, 2022, the entire contents of each of which incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a refrigeration cycle system including a cascade heat exchanger that exchanges heat between a first refrigerant circulating in a first refrigerant circuit and a second refrigerant that is a CO.sub.2 refrigerant circulating in a second refrigerant circuit.

BACKGROUND ART

[0003] In recent years, in order to reduce power consumption in the whole society, it is required to reduce power consumption of an air conditioner and the like. For example, Patent Literature 1 (JP 2002-147819 A) discloses an air conditioner that is a refrigeration apparatus capable of reliably reducing power consumption in response to a request for power peak cut. In the configuration of the refrigeration apparatus in Patent Literature 1, a controller controls a capacity of a compressor unit such that a detected current value is within a range that does not exceed a predetermined set value.

SUMMARY

[0004] A refrigeration cycle system according to a first aspect includes a first refrigerant circuit, a second refrigerant circuit, and a controller. The first refrigerant circuit includes a first compressor that circulates a first refrigerant at a discharge refrigerant pressure in a range of 0.5 MPa or more and 4 MPa or less, and a cascade heat exchanger that cools a second refrigerant that is a CO.sub.2 refrigerant by the first refrigerant, and implements a first vapor compression refrigeration cycle using the first refrigerant. The second refrigerant circuit includes a second compressor that circulates the second refrigerant at a discharge refrigerant pressure in a range of 5 MPa or more and 14 MPa or less, and implements a second vapor compression refrigeration cycle using the second refrigerant. The controller controls the first compressor and the second compressor. The controller controls the first compressor and the second compressor such that an intermediate temperature between an evaporation temperature of the first refrigerant and a condensation temperature of the second refrigerant in the cascade heat exchanger satisfies $T2 + (T1 - T2) \times 0.1 \leq TV \leq T2 + (T1 - T2) \times 0.4$, where $T1^{\circ} \text{C.}$ represents a pressure equivalent saturation temperature of a refrigerant discharged from the first compressor, and $T2^{\circ} \text{C.}$ represents a pressure equivalent saturation temperature of a refrigerant sucked into the second compressor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. **1** is a circuit diagram showing a schematic configuration of a refrigeration cycle system according to an embodiment.

[0006] FIG. **2** is a circuit diagram of a refrigeration cycle system in which a flow direction of a refrigerant during a cooling operation is added to FIG. **1**.

[0007] FIG. **3** is a circuit diagram of a refrigeration cycle system in which a flow direction of a refrigerant during a heating operation is added to FIG. **1**.

[0008] FIG. **4** is a conceptual diagram using a Mollier diagram for describing heat exchange in a cascade heat exchanger.

[0009] FIG. **5** is a graph showing a relationship between an intermediate temperature of the cascade heat exchanger and a total power consumption of the refrigeration cycle system.

[0010] FIG. **6** is a diagram for describing a relationship between a value obtained by normalizing the intermediate temperature of the cascade heat exchanger and the intermediate temperature.

[0011] FIG. **7** is a graph showing a relationship between rotational speeds of a first compressor and a second compressor and compressor efficiency.

DESCRIPTION OF EMBODIMENTS

(1) Overall Configuration

[0012] A refrigeration cycle system **1** according to an embodiment has a configuration shown in FIG. **1**. The refrigeration cycle system **1** includes a first refrigerant circuit **100** through which a first refrigerant circulates and a second refrigerant circuit **200** through which a carbon dioxide refrigerant (CO.sub.2 refrigerant) circulates. The refrigeration cycle system **1** is a device that performs indoor air conditioning by exchanging heat between a carbon dioxide refrigerant as a heat transfer medium and indoor air. Examples of the indoor air conditioning include cooling and heating.

(1-1) First Refrigerant Circuit **100**

[0013] The first refrigerant circuit **100** includes a first compressor **110**, a first four-way valve **120**, an outdoor air heat exchanger **130**, and a cascade heat exchanger **150**. As the first refrigerant used in the first refrigerant circuit **100**, for example, R32, R454C, propane, R1234yf, R1234ze, or ammonia can be used. Alternatively, a refrigerant including any of R32, R454C, propane, R1234yf, R1234ze, or ammonia can be used as the first refrigerant. The outdoor air heat exchanger **130** is an example of an outdoor heat exchanger.

[0014] The first compressor **110** is a device that compresses the first refrigerant in a gaseous state. As the first compressor **110**, for example, a positive displacement compressor can be used. Examples of the positive displacement compressor include a rotary compressor and a scroll compressor. The first compressor **110** is driven by, for example, a motor **115**. The motor **115** is, for example, a motor including an inverter and capable of changing a rotational speed. By changing the rotational speed of the motor **115**, the first compressor **110** can adjust a discharge capacity.

[0015] The first four-way valve **120** is a device for switching a flow of the first refrigerant in first refrigerant circuit **100**. Specifically, the first four-way valve **120** switches between a communication state indicated by a solid line of the first four-way valve **120** in FIG. **1** and a communication state indicated by a broken line of the first four-way valve **120** in FIG. **1**. For example, the first four-way valve **120** can be replaced with another configuration having a function of switching between a first refrigerant heat radiation state and a first refrigerant evaporation state by combining a plurality of valves (for example, electromagnetic valves or three-way valves) other than four-way valves.

[0016] When the first four-way valve **120** is switched to the communication state indicated by the solid line, the first refrigerant circuit **100** enters the first refrigerant heat radiation state in which the outdoor air heat exchanger **130** functions as a radiator for the first refrigerant and the cascade heat exchanger **150** functions as an evaporator for the first refrigerant. The first refrigerant heat

radiation state is a state in which the first refrigerant is cooled by outside air (outdoor air). When the first four-way valve **120** is switched to the communication state indicated by the broken line, the first refrigerant circuit **100** enters the first refrigerant evaporation state in which the outdoor air heat exchanger **130** functions as an evaporator for the first refrigerant and the cascade heat exchanger **150** functions as a radiator for the first refrigerant. The first refrigerant evaporation state is a state in which the first refrigerant is heated by outside air.

[0017] In the first four-way valve **120**, in the first refrigerant heat radiation state, a discharge port of the first compressor **110** communicates with the outdoor air heat exchanger **130**, and a suction port of the first compressor **110** communicates with the cascade heat exchanger **150**. In the first four-way valve **120**, in the first refrigerant evaporation state, the discharge port of the first compressor **110** communicates with the cascade heat exchanger **150**, and the suction port of the first compressor **110** communicates with the outdoor air heat exchanger **130**.

[0018] The outdoor air heat exchanger **130** is a device that exchanges heat between the first refrigerant and outside air. The outdoor air heat exchanger **130** is, for example, a fin-and-tube heat exchanger. In a state where the first four-way valve **120** is switched to the first refrigerant heat radiation state (state indicated by the solid line), the outdoor air heat exchanger **130** functions as a radiator for the first refrigerant using the outside air as a cooling source. In a state where the first four-way valve **120** is switched to the first refrigerant evaporation state (state indicated by the broken line), the outdoor air heat exchanger **130** functions as an evaporator for the first refrigerant using the outside air as a heating source. The outdoor air heat exchanger **130** communicates with the first four-way valve **120** and a first expansion valve **140**.

[0019] The first refrigerant circuit **100** includes the first expansion valve **140**. The first expansion valve **140** is a device that decompresses the first refrigerant. The first expansion valve **140** is a device for adjusting the pressure of the first refrigerant having passed or adjusting the flow rate of the refrigerant. The first expansion valve **140** is, for example, an electric expansion valve that changes an opening degree in accordance with an electric signal. The first expansion valve **140** is provided between the outdoor air heat exchanger **130** and the cascade heat exchanger **150**. The first expansion valve **140** adjusts the pressure of the first refrigerant flowing between the outdoor air heat exchanger **130** and the cascade heat exchanger **150**. In a state where the first four-way valve **120** is switched to the first refrigerant heat radiation state, the first expansion valve **140** decompresses the first refrigerant that has radiated heat in the outdoor air heat exchanger **130**. In a state where the first four-way valve **120** is switched to the first refrigerant evaporation state, the first expansion valve **140** decompresses the first refrigerant having radiated heat in the cascade heat exchanger **150**. Note that, the first expansion valve **140** is not limited to the electric expansion valve. As the first expansion valve **140**, for example, an expansion valve or a capillary tube of a type other than the electric expansion valve may be used.

[0020] The cascade heat exchanger **150** is a device that exchanges heat between the first refrigerant and the carbon dioxide refrigerant. As the cascade heat exchanger **150**, for example, a plate heat exchanger, a double tube heat exchanger, or a shell-and-tube heat exchanger can be used. In a state where the first four-way valve **120** is switched to the first refrigerant heat radiation state, the cascade heat exchanger **150** functions as an evaporator for the first refrigerant using the carbon dioxide refrigerant as a heating source. In a state where the first four-way valve **120** is switched to the first refrigerant evaporation state, the cascade heat exchanger **150** functions as a radiator for the first refrigerant using the carbon dioxide refrigerant as a cooling source. The cascade heat exchanger **150** communicates with the first four-way valve **120**, and communicates with the outdoor air heat exchanger **130** via the first expansion valve **140**.

(1-2) Second Refrigerant Circuit **200**

[0021] The second refrigerant circuit **200** includes a second compressor **210**, a cascade heat exchanger **150**, a second four-way valve **220**, and a plurality of indoor air heat exchangers **251**, **252**, and **253**. The indoor air heat exchangers **251**, **252**, and **253** are an example of indoor heat

exchangers. In the second refrigerant circuit **200**, the carbon dioxide refrigerant circulates as the heat transfer medium. In this example, the second refrigerant circuit **200** includes three indoor air heat exchangers **251**, **252**, and **253**. However, the number of indoor air heat exchangers may be one, two, or four or more. The cascade heat exchanger **150** is shared by two circuits, namely, the first refrigerant circuit **100** and the second refrigerant circuit **200**.

[0022] The second compressor **210** is a device that compresses the carbon dioxide refrigerant in a gaseous state. As the second compressor **210**, for example, a positive displacement compressor can be used. Examples of the positive displacement compressor include a rotary compressor and a scroll compressor. The second compressor **210** is driven by, for example, a motor **215**. The motor **215** is, for example, a motor including an inverter and capable of changing a rotational speed. By changing the rotational speed of the motor **215**, the second compressor **210** can adjust a discharge capacity.

[0023] The second four-way valve **220** is a device for switching the flow of the carbon dioxide refrigerant in the second refrigerant circuit **200**. Specifically, the second four-way valve **220** switches between a communication state indicated by a solid line of the second four-way valve **220** in FIG. 1 and a communication state indicated by a broken line of the second four-way valve **220** in FIG. 1. For example, the second four-way valve **220** can be replaced with another configuration having a function of switching between a CO.sub.2 heat radiation state and a CO.sub.2 evaporation state by combining a plurality of valves (for example, electromagnetic valves or three-way valves).

[0024] When the second four-way valve **220** is switched to the communication state indicated by the solid line, the second refrigerant circuit **200** enters the CO.sub.2 evaporation state in which the indoor air heat exchangers **251**, **252**, and **253** function as an evaporator for the carbon dioxide refrigerant and the cascade heat exchanger **150** functions as a radiator for the carbon dioxide refrigerant. The CO.sub.2 evaporation state is a state in which the carbon dioxide refrigerant evaporates to cool the indoor air. When the second four-way valve **220** is switched to a communication state indicated by the broken line, the second refrigerant circuit **200** enters the CO.sub.2 heat radiation state in which the indoor air heat exchangers **251**, **252**, and **253** function as a radiator for the carbon dioxide refrigerant and the cascade heat exchanger **150** functions as an evaporator for the carbon dioxide refrigerant. The CO.sub.2 heat radiation state is a state in which the carbon dioxide refrigerant radiates heat to indoor air.

[0025] In the second four-way valve **220**, in the CO.sub.2 heat radiation state, a discharge port of the second compressor **210** communicates with the indoor air heat exchangers **251**, **252**, and **253**, and a suction port of the second compressor **210** communicates with the cascade heat exchanger **150**. In the second four-way valve **220**, in the CO.sub.2 evaporation state, the discharge port of the second compressor **210** communicates with the cascade heat exchanger **150**, and the suction port of the second compressor **210** communicates with the indoor air heat exchangers **251**, **252**, and **253**.

[0026] Each of the indoor air heat exchangers **251**, **252**, and **253** is, for example, a device that exchanges heat between indoor air in a corresponding room and the carbon dioxide refrigerant. The indoor air heat exchangers **251**, **252**, and **253** are, for example, fin-and-tube heat exchangers. The indoor air heat exchangers **251**, **252**, and **253** function as a radiator for the carbon dioxide refrigerant in order to heat the indoor air in a state where the second four-way valve **220** is switched to the CO.sub.2 heat radiation state. In other words, the indoor air heat exchangers **251**, **252**, and **253** function as a radiator of a carbon dioxide refrigerant using the indoor air as a cooling source in the CO.sub.2 heat radiation state. In a state where the second four-way valve **220** is switched to the CO.sub.2 evaporation state, the indoor air heat exchangers **251**, **252**, and **253** function as an evaporator for the carbon dioxide refrigerant in order to cool the indoor air. In other words, the indoor air heat exchangers **251**, **252**, and **253** function as an evaporator for the carbon dioxide refrigerant using the indoor air as a heating source in the CO.sub.2 evaporation state. In the CO.sub.2 evaporation state, the indoor air heat exchangers **251**, **252**, and **253** communicate with the suction port of the second compressor **210** via the second four-way valve **220**, and

communicate with the cascade heat exchanger **150** via corresponding second expansion valves **261**, **262**, and **263**, respectively, and a third expansion valve **230**.

[0027] The second refrigerant circuit **200** includes the second expansion valves **261**, **262**, and **263** and the third expansion valve **230**. The second expansion valves **261**, **262**, and **263** and the third expansion valve **230** are devices for adjusting the pressure of the carbon dioxide refrigerant having passed and adjusting the flow rate of the carbon dioxide refrigerant. The second expansion valves **261**, **262**, and **263** and the third expansion valve **230** are, for example, electric expansion valves that change the opening degree in accordance with an electric signal. The third expansion valve **230** is connected to the cascade heat exchanger **150**. The third expansion valve **230** is connected to all the second expansion valves **261**, **262**, and **263**. The second expansion valves **261**, **262**, and **263** are connected to the corresponding indoor air heat exchangers **251**, **252**, and **253**, respectively.

[0028] The second expansion valves **261**, **262**, and **263** are provided between the corresponding indoor air heat exchangers **251**, **252**, and **253** and the cascade heat exchanger **150**. The second expansion valves **261**, **262**, and **263** adjust the pressure of the carbon dioxide refrigerant flowing through the corresponding indoor air heat exchangers **251**, **252**, and **253**. The second expansion valves **261**, **262**, and **263** decompress the carbon dioxide refrigerant that has radiated heat in the indoor air heat exchangers **251**, **252**, and **253** in a state where the second four-way valve **220** is switched to the CO.sub.2 heat radiation state. The second expansion valves **261**, **262**, and **263** decompress the carbon dioxide refrigerant to be sent to the indoor air heat exchangers **251**, **252**, and **253** in a state where the second four-way valve **220** is switched to the CO.sub.2 evaporation state.

[0029] The third expansion valve **230** is provided between all the indoor air heat exchangers **251**, **252**, and **253** and the cascade heat exchanger **150**. The third expansion valve **230** adjusts the pressure of the carbon dioxide refrigerant flowing from the indoor air heat exchangers **251**, **252**, and **253** to the cascade heat exchanger **150**. In a state where second four-way valve **220** is switched to the CO.sub.2 evaporation state, the third expansion valve **230** is fully opened so as not to decompress the carbon dioxide refrigerant that has radiated heat in the cascade heat exchanger **150** as much as possible. The third expansion valve **230** decompresses the carbon dioxide refrigerant that has radiated heat in the indoor air heat exchangers **251**, **252**, and **253** in a state where the second four-way valve **220** is switched to the CO.sub.2 heat radiation state. However, the third expansion valve **230** is not limited to the electric expansion valve. As the third expansion valve **230**, for example, an expansion valve or a capillary tube of a type other than the electric expansion valve may be used.

[0030] The cascade heat exchanger **150** functions as a radiator for the carbon dioxide refrigerant using the first refrigerant as a cooling source in a state where the first four-way valve **120** is switched to the first refrigerant heat radiation state (state indicated by the solid line) and the second four-way valve **220** is switched to the CO.sub.2 evaporation state (state indicated by the solid line). The cascade heat exchanger **150** functions as an evaporator for the carbon dioxide refrigerant using the first refrigerant as a heating source in a state where first four-way valve **120** is switched to the first refrigerant evaporation state (state indicated by the broken line) and the second four-way valve **220** is switched to the CO.sub.2 heat radiation state (state indicated by the broken line). The cascade heat exchanger **150** communicates with the second four-way valve **220** and communicates with the indoor air heat exchangers **251**, **252**, and **253** via the third expansion valve **230** and the second expansion valves **261**, **262**, and **263** in order to cause the carbon dioxide refrigerant to flow.

[0031] The constituent devices of the first refrigerant circuit **100** and the second refrigerant circuit **200** are provided in a heat transfer unit **2** and a plurality of utilization units **5a**, **5b**, and **5c**. The utilization units **5a**, **5b**, and **5c** are provided corresponding to the indoor air heat exchangers **251**, **252**, and **253**, respectively.

[0032] The heat transfer unit **2** is disposed outdoors. The cascade heat exchanger **150**, the second compressor **210**, the third expansion valve **230**, and the second four-way valve **220** are provided in

the heat transfer unit **2**. The heat transfer unit **2** is provided with an outdoor fan **160** that supplies outside air to the outdoor air heat exchanger **130**. The outdoor fan **160** is driven by, for example, a motor **165** that can change a rotational speed. The outdoor fan **160** is, for example, a propeller fan. The outdoor fan **160** can change the volume of airflow passing through the outdoor air heat exchanger **130** by changing the rotational speed of the motor **165**.

[0033] The utilization units **5a**, **5b**, and **5c** are disposed indoors. The indoor air heat exchangers **251**, **252**, and **253** of the second refrigerant circuit **200** are accommodated in the utilization units **5a**, **5b**, and **5c**, respectively. The second expansion valves **261**, **262**, and **263** of the second refrigerant circuit **200** are also accommodated in the utilization units **5a**, **5b**, and **5c**, respectively. The utilization units **5a**, **5b**, and **5c** accommodate indoor fans **271**, **272**, and **273** that supplies indoor air to the indoor air heat exchangers **251**, **252**, and **253**, respectively. The indoor fans **271**, **272**, and **273** are, for example, centrifugal fans or multiblade fans. The indoor fans **271**, **272**, and **273** are driven by motors **276**, **277**, and **278**, respectively. Each of the motors **276**, **277**, and **278** is, for example, a motor including an inverter and capable of changing a rotational speed. By changing the rotational speed of the motor **276**, **277**, and **278**, the indoor fans **271**, **272**, and **273** can change the volume of airflow passing through the indoor air heat exchangers **251**, **252**, and **253**.

[0034] The heat transfer unit **2** and the utilization units **5a**, **5b**, and **5c** are connected by connection pipes **6** and **7** constituting a part of the second refrigerant circuit **200**. The connection pipe **6** is a pipe for allowing the cascade heat exchanger **150** and one end of the second expansion valves **261**, **262**, and **263** to communicate with each other. The connection pipe **7** is a pipe for allowing the second four-way valve **220** and the indoor air heat exchangers **251**, **252**, and **253** to communicate with each other.

[0035] The constituent devices of the heat transfer unit **2** and the utilization units **5a**, **5b**, and **5c** are controlled by a controller **300**. The controller **300** is configured by, for example, communicably connecting control boards (not shown) provided in the heat transfer unit **2** and the utilization units **5a**, **5b**, and **5c**. In FIG. **1**, for convenience, the controller **300** is shown at a position away from the heat transfer unit **2** and the utilization units **5a**, **5b**, and **5c**. The controller **300** controls the constituent devices of the refrigeration cycle system **1** (here, the heat transfer unit **2** and the utilization units **5a**, **5b**, and **5c**). In other words, the controller **300** controls the operation of the entire refrigeration cycle system **1**.

[0036] The controller **300** is achieved by a computer. The controller **300** includes a control calculation device and a storage device. A processor such as a CPU or a GPU can be used for the control calculation device. The control calculation device reads a program stored in the storage device, and executes predetermined image processing and calculation processing in accordance with this program. Furthermore, the control calculation device can write a calculation result to the storage device and read information stored in the storage device in accordance with the program.

[0037] The refrigeration cycle system **1** includes the heat transfer unit **2**, the plurality of utilization units **5a**, **5b**, and **5c** connected in parallel to each another, the connection pipes **6** and **7** connecting the heat transfer unit **2** and the utilization units **5a**, **5b**, and **5c**, and the controller **300** that controls the constituent devices of the heat transfer unit **2** and the utilization units **5a**, **5b**, and **5c**.

(2) Behavior of Refrigeration Cycle System **1**

[0038] Next, a behavior of the refrigeration cycle system **1** is described with reference to FIGS. **1** to **3**. FIG. **2** shows a behavior (flow of the first refrigerant and the carbon dioxide refrigerant) in the cooling operation of the refrigeration cycle system **1**. FIG. **3** shows a behavior in the heating operation. The refrigeration cycle system **1** can perform the cooling operation for cooling indoor air and the heating operation for heating indoor air for indoor air conditioning. In the cooling operation and the heating operation, the behavior of the refrigeration cycle system **1** is controlled by the controller **300**.

(2-1) Cooling Operation

[0039] When all of the utilization units **5a**, **5b**, and **5c** perform the cooling operation during the

cooling operation, for example, the first four-way valve **120** is switched to the first refrigerant heat radiation state (the first four-way valve **120** is in the state indicated by the solid line), and the second four-way valve **220** is switched to the CO.sub.2 evaporation state (the second four-way valve **220** is in the state indicated by the solid line).

[0040] In the first refrigerant circuit **100**, the first refrigerant discharged from the first compressor **110** is sent to the outdoor air heat exchanger **130** through the first four-way valve **120**. The first refrigerant sent to the outdoor air heat exchanger **130** exchanges heat with the outside air supplied by the outdoor fan **160** and is cooled to be condensed. The first refrigerant having radiated heat in the outdoor air heat exchanger **130** is decompressed by the first expansion valve **140** and then sent to the cascade heat exchanger **150**. The first refrigerant sent to the cascade heat exchanger **150** is heated and evaporated by exchanging heat with the carbon dioxide refrigerant in the cascade heat exchanger **150** functioning as an evaporator for the first refrigerant. The first refrigerant evaporated in the cascade heat exchanger **150** is sucked into the first compressor **110** through the first four-way valve **120**, and is discharged again from the first compressor **110**.

[0041] In the second refrigerant circuit **200**, the carbon dioxide refrigerant discharged from the second compressor **210** is sent to the cascade heat exchanger **150** through the second four-way valve **220**. The carbon dioxide refrigerant sent to the cascade heat exchanger **150** exchanges heat with the first refrigerant in the cascade heat exchanger **150** functioning as a radiator for the carbon dioxide refrigerant to be cooled. The carbon dioxide refrigerant having radiated heat in the cascade heat exchanger **150** passes through the third expansion valve **230** in a fully opened state, and then is sent to the connection pipe **6** to be branched. The carbon dioxide refrigerant branched at the connection pipe **6** is decompressed by the second expansion valves **261**, **262**, and **263** and then sent to the indoor air heat exchangers **251**, **252**, and **253**. The carbon dioxide refrigerant sent to the indoor air heat exchangers **251**, **252**, and **253** exchanges heat with indoor air supplied by the indoor fans **271**, **272**, and **273** and is heated to be evaporated. In each of the utilization units **5a**, **5b**, and **5c**, the cooling operation for cooling indoor air is performed. The carbon dioxide refrigerant evaporated in the indoor air heat exchangers **251**, **252**, and **253** merges at the connection pipe **7**. The carbon dioxide refrigerant having merged at the connection pipe **7** is sucked into the second compressor **210** through the second four-way valve **220**, and is discharged again from the second compressor **210**.

[0042] In the cooling operation, the first refrigerant circuit **100** executes a first vapor compression refrigeration cycle using the first refrigerant. In the cooling operation, the second refrigerant circuit **200** executes a second vapor compression refrigeration cycle using the carbon dioxide refrigerant as the second refrigerant.

(2-1-1) Control in Cooling Operation

[0043] The first compressor **110** circulates the first refrigerant in the first refrigerant circuit **100** at a discharge refrigerant pressure in a range of 0.5 [MPa] or more and to 4 [MPa] or less. The second compressor **210** circulates the carbon dioxide refrigerant at a discharge refrigerant pressure in a range of 5 [MPa] or more and 14 [MPa] or less. During the cooling operation, the second refrigerant circuit **200** operates such that the carbon dioxide refrigerant is condensed by the cascade heat exchanger **150**.

[0044] In FIG. **4**, a Mollier diagram M**100** of the first refrigerant circuit **100** and a Mollier diagram M**200** of the second refrigerant circuit **200** are illustrated by overlapping graphs having different vertical axes. FIG. **4** shows an image to describe heat exchange in the cascade heat exchanger **150**, and thus is not a detailed drawing. The first refrigerant circuit **100** operates such that a pressure (condensation pressure PH**1**) of the first refrigerant discharged from the first compressor **110** satisfies a condition of $0.5 \text{ [MPa]} \leq \text{PH1} \leq 4 \text{ [MPa]}$. The second refrigerant circuit **200** operates such that a pressure (evaporation pressure PL**1**) of the carbon dioxide refrigerant discharged from the second compressor **210** satisfies a condition of $5 \text{ [MPa]} \leq \text{PH2} \leq 14 \text{ [MPa]}$. A portion surrounded by a square frame line in FIG. **4** indicates a place where heat is exchanged between the carbon dioxide

refrigerant in a gaseous state of high temperature and high pressure and the first refrigerant in a liquid state of low temperature and low pressure or a gas-liquid two-phase state in the cascade heat exchanger **150**.

[0045] The controller **300** controls the first compressor **110** and the second compressor **210** such that a target value TV [° C.] of an intermediate temperature T_m between an evaporation temperature T_e of the first refrigerant and a condensation temperature T_c of the carbon dioxide refrigerant in the cascade heat exchanger **150** satisfies the following formula (1), where T₁ [° C.] represents a pressure equivalent saturation temperature of the refrigerant discharged from first compressor **110**, and T₂ [° C.] represents a pressure equivalent saturation temperature of the refrigerant sucked into the second compressor **210**.

$$T_2 + (T_1 - T_2) \times 0.1 \leq TV \leq T_2 + (T_1 - T_2) \times 0.4 \quad (1)$$

In order to make the intermediate temperature T_m [° C.] coincide with the target value TV [° C.], the controller **300** adjusts the rotational speeds of the first compressor **110** and the second compressor **210**, for example. The intermediate temperature T_m is a temperature between the evaporation temperature T_e of the first refrigerant and the condensation temperature T_c of the carbon dioxide refrigerant. The intermediate temperature T_m is given by, for example, a formula of T_m=(T_e+T_c)/2 from the target evaporation temperature T_e of the first refrigerant and the condensation temperature T_c of the carbon dioxide refrigerant. For example, as compared with a case where the target value TV is set to {T₂+(T₁-T₂)×0.5}, setting the target value TV to {T₂+(T₁-T₂)×0.3} increases the rotational speed of the first compressor **110** under the same other conditions. In other words, as compared with the case of TV=T₂+(T₁-T₂)×0.5, when TV=T₂+(T₁-T₂)×0.3, a load on the first compressor **110** increases and a load on the second compressor **210** decreases.

[0046] Here, a case where the controller **300** performs control such that the intermediate temperature T_m coincides with the target value TV will be described. However, a method of controlling the first compressor **110** and the second compressor **210** such that the intermediate temperature T_m becomes the target value TV is not limited to such a method. For example, the controller **300** can be configured to control the first compressor **110** and the second compressor **210** such that the intermediate temperature T_m does not deviate from a predetermined range (for example, a range of {T₂+(T₁-T₂)×0.1} or more and {T₂+(T₁-T₂)×0.4} or less) of the target value TV. In addition, for example, a plurality of the target values TV may be prepared and switched in accordance with a situation (for example, outside air temperature).

[0047] In the refrigeration cycle system **1**, the controller **300** performs control similar to conventional control except that the first compressor **110** and the second compressor **210** are controlled such that intermediate temperature T_m in the cascade heat exchanger **150** becomes the target value TV.

[0048] In FIG. 5, the horizontal axis represents the intermediate temperature T_m when the pressure equivalent saturation temperature T₁ of the refrigerant discharged from the first compressor **110** is 45 [° C.] and the pressure equivalent saturation temperature T₂ of the refrigerant sucked into the second compressor **210** is 8 [° C.]. FIG. 5 shows a relationship between the intermediate temperature T_m and a total power consumption of the refrigeration cycle system **1**. Here, the total power consumption [KW] of the refrigeration cycle system **1** is a power consumption of the refrigeration cycle system **1**, and is the sum of a power consumption [KW] of the first refrigerant circuit **100** and a power consumption [KW] of the second refrigerant circuit. In FIG. 5, the vertical axis represents power consumption necessary for obtaining predetermined capability in the cascade heat exchanger **150**. It can be understood from FIG. 5 that the graph indicating the relationship between the intermediate temperature T_m of the first refrigerant circuit **100** and the total power consumption of the refrigeration cycle system **1** has a downward convex shape, and that there is a range in which the power consumption can be reduced. It can be understood from FIG. 5 that it is

preferable to set the target value TV of the intermediate temperature Tm within a range where the total power consumption is 9.5 [KW] or less. Furthermore, in the case as shown in FIG. 5, it is more preferable to set the target value TV of the intermediate temperature Tm within a range where the total power consumption is 9.4 [KW] or less. The tendency shown in FIG. 5 occurs by employing a first compressor that circulates the first refrigerant at the discharge refrigerant pressure in a range of 0.5 MPa or more and 4 MPa or less and the second compressor that circulates the second refrigerant at the discharge refrigerant pressure in a range of 5 MPa or more and 14 MPa or less.

[0049] FIG. 6 shows a relationship between a value obtained by normalizing the intermediate temperature Tm and the intermediate temperature. FIG. 6 shows a value obtained by normalizing the intermediate temperature when the pressure equivalent saturation temperature 8 [° C.] of the refrigerant sucked into the second compressor **210** is “0” and the pressure equivalent saturation temperature 45 [° C.] of the refrigerant discharged from the first compressor **110** is “1”. A range (range of the target value) in which the normalized value is 0.1 or more and 0.4 or less is indicated by an arrow in FIG. 5. It can be understood from FIG. 5 that the total power consumption in the range indicated by the arrow represents a low value. In other words, when the first compressor **110** that circulates the first refrigerant at the discharge refrigerant pressure in the range of 0.5 MPa or more and 4 MPa or less and the second compressor **210** that circulates the second refrigerant at the discharge refrigerant pressure in the range of 5 MPa or more and 14 MPa or less are employed, the target value TV satisfies $T2+(T1-T2) \times 0.1 \leq TV \leq T2+(T1-T2) \times 0.4$, and thus, the total power consumption represents a low value. This normalized value is an appropriate range of a constant a when expressed as $TV = T2 + (T1 - T2) \times a$. When a normalized value corresponding to the target value TV is set in the range of the intermediate temperature Tm at which the total power consumption is 9.4 [kW] or less, the normalized value is preferably 0.15 or more and 0.35 or less. In other words, when it is desired to further reduce the total power consumption with the use of the first compressor **110** and the second compressor **210** as described above, the target value TV preferably satisfies $T2+(T1-T2) \times 0.15 \leq TV \leq T2+(T1-T2) \times 0.35$.

[0050] The controller **300** may control the first compressor **110** and the second compressor **210** such that the intermediate temperature Tm in the cascade heat exchanger **150** satisfies the following formula (2).

$$T2+(T1-T2) \times 0.1 \leq Tm \leq T2+(T1-T2) \times 0.4 \quad (2)$$

[0051] As a method of controlling the first compressor **110** and the second compressor **210** so that the intermediate temperature Tm satisfies the above formula (2), for example, there is a method of controlling the evaporation temperature Te of the first refrigerant and the condensation temperature Tc of the carbon dioxide refrigerant in the cascade heat exchanger **150**.

[0052] Specifically, a target evaporation temperature Tme of evaporation temperature Te of the first refrigerant in the cascade heat exchanger **150** is set, the first compressor **110** is controlled to reach the target evaporation temperature Tme, and a target condensation temperature Tmc of the condensation temperature Tc of the carbon dioxide refrigerant is set, and the second compressor **210** is controlled to reach the target condensation temperature Tmc.

[0053] The target condensation temperature Tmc and the target evaporation temperature Tme are set to values within a range where the intermediate temperature Tm satisfies the above formula (2) on the basis of the total power consumption of the refrigeration cycle system **1** during the operation of the first compressor **110** and the second compressor **210**.

[0054] FIG. 7 shows a relationship between the rotational speed and the compressor efficiency of each of the first compressor **110** and the second compressor **210**. In the example shown in FIG. 7, both the first compressor **110** and the second compressor **210** have the highest compressor efficiency near a rotational speed R. R is a value between a lower limit value Rmin and an upper limit value Rmax. In both the first compressor **110** and the second compressor **210**, the compressor

efficiency gradually decreases as the rotational speed becomes smaller than near R , and becomes the lowest at the lower limit value R_{min} of the rotational speed. In both the first compressor **110** and the second compressor **210**, the compressor efficiency decreases as the rotational speed becomes larger than near R , and the compressor efficiency at the upper limit value R_{max} of the rotational speed is lower than the compressor efficiency near the rotational speed R . When the compressor efficiency of the first compressor **110** is compared with the compressor efficiency of the second compressor **210**, the compressor efficiency of the second compressor **210** is lower than the compressor efficiency of the first compressor **110** by about several percent in an entire range from the lower limit value R_{min} of the rotational speed to the upper limit value R_{max} of the rotational speed used in the refrigeration cycle system **1**. This tendency is caused by the fact that a discharge pressure of the second compressor **210** is higher than a discharge pressure of the first compressor **110**.

[0055] In order for the controller **300** to control the intermediate temperature T_m , the refrigeration cycle system **1** includes a high-pressure pressure sensor **410** on a discharge side of the first compressor **110** for detecting the pressure equivalent saturation temperature T_1 of the refrigerant discharged from the first compressor **110**. The refrigeration cycle system **1** includes a low-pressure pressure sensor **420** on a suction side of the second compressor **210** for detecting the pressure equivalent saturation temperature T_2 of the refrigerant sucked into the second compressor **210**. A low-pressure pressure sensor **430** for detecting the evaporation temperature T_e of the cascade heat exchanger **150** is provided on a suction side of the first compressor **110**. Furthermore, a high-pressure pressure sensor **440** for detecting the condensation temperature T_c of the cascade heat exchanger **150** is provided on a discharge side of the second compressor **210**. The controller **300** of the refrigeration cycle system **1** uses detection results of various sensors other than the pressure sensors **410**, **420**, **430**, and **440**. However, sensors similar to the conventional sensors may be used, and thus sensors other than the pressure sensors **410**, **420**, **430**, and **440** are not described here.

(2-2) Heating Operation

[0056] When all of the utilization units **5a**, **5b**, and **5c** perform the heating operation during the heating operation, for example, the first four-way valve **120** is switched to the first refrigerant evaporation state (the first four-way valve **120** is in the state indicated by the broken line), and the second four-way valve **220** is switched to the CO.sub.2 heat radiation state (the second four-way valve **220** is in the state indicated by the broken line).

[0057] In the first refrigerant circuit **100**, the first refrigerant discharged from the first compressor **110** is sent to the cascade heat exchanger **150** through the first four-way valve **120**. The first refrigerant sent to the cascade heat exchanger **150** exchanges heat with the carbon dioxide refrigerant and is cooled to be condensed. The first refrigerant having radiated heat in the cascade heat exchanger **150** is decompressed by the first expansion valve **140** and then sent to the outdoor air heat exchanger **130**. The first refrigerant sent to the outdoor air heat exchanger **130** exchanges heat with the outside air supplied by the outdoor fan **160** in the outdoor air heat exchanger **130** to be heated and evaporated. The first refrigerant evaporated in the outdoor air heat exchanger **130** is sucked into the first compressor **110** through the first four-way valve **120**, and is discharged again from the first compressor **110**.

[0058] In the second refrigerant circuit **200**, the carbon dioxide refrigerant discharged from the second compressor **210** passes through the second four-way valve **220**, and then, is branched at the connection pipe **7**. The carbon dioxide refrigerant branched at the connection pipe **7** is sent to the indoor air heat exchangers **251**, **252**, and **253**. The carbon dioxide refrigerant sent to the indoor air heat exchangers **251**, **252**, and **253** exchanges heat with indoor air supplied by the indoor fans **271**, **272**, and **273** and is cooled. In each of the utilization units **5a**, **5b**, and **5c**, the heating operation for heating indoor air is performed. The carbon dioxide refrigerant having radiated heat in the indoor air heat exchangers **251**, **252**, and **253** is decompressed by the second expansion valves **261**, **262**, and **263** and then joins at the connection pipe. The carbon dioxide refrigerant joined at the

connection pipe **6** is further decompressed by the third expansion valve **230** and then sent to the cascade heat exchanger **150**. The carbon dioxide refrigerant sent to the cascade heat exchanger **150** exchanges heat with the first refrigerant to be heated. The carbon dioxide refrigerant that has passed through the cascade heat exchanger **150** is sucked into the second compressor **210** through the second four-way valve **220**, and is discharged again from the second compressor **210**.

[0059] In the heating operation, the first refrigerant circuit **100** executes a first vapor compression refrigeration cycle using the first refrigerant. In a heating cooling operation, the second refrigerant circuit **200** executes a second vapor compression refrigeration cycle using the carbon dioxide refrigerant as the second refrigerant.

(3) Characteristics

[0060] In the refrigeration cycle system **1** according to the embodiment, the controller **300** controls the first compressor **110** such that the intermediate temperature T_m in the cascade heat exchanger **150** becomes the target value TV that satisfies $T_2 + (T_1 - T_2) \times 0.1 \leq TV \leq T_2 + (T_1 - T_2) \times 0.4$. In the above formula, T_1 [° C.] is the pressure equivalent saturation temperature of the refrigerant discharged from the first compressor **110**, and T_2 [° C.] is the pressure equivalent saturation temperature of the refrigerant sucked into the second compressor **210**. In order to further reduce power consumption, it is preferable to control the first compressor **110** and the second compressor **210** such that the intermediate temperature T_m satisfies $T_2 + (T_1 - T_2) \times 0.15 \leq TV \leq T_2 + (T_1 - T_2) \times 0.35$.

[0061] By performing such control by the controller **300**, the amount of heat can be increased in the first refrigerant circuit **100** including the first compressor **110** having high efficiency, and the power consumption of the entire refrigeration cycle system **1** can be reduced.

(4) Modifications

(4-1) A

[0062] In the above embodiment, a case has been described where the pressure sensors **410** to **440** are used to detect the pressure equivalent saturation temperature, the evaporation temperature, and the condensation temperature. However, a detection device used to detect the temperatures is not limited to a pressure sensor. Examples of the detection device other than a pressure sensor include a temperature sensor.

(4-2) B

[0063] In the embodiment, a case has been described where the constituent devices of the first refrigerant circuit **100** and the second refrigerant circuit **200** are provided in the heat transfer unit **2** and the plurality of utilization units **5a**, **5b**, and **5c**. However, the cascade heat exchanger **150** and the second compressor **210** of the second refrigerant circuit **200** may be provided in a separate cascade unit instead of the heat transfer unit and the utilization unit.

(5) Other

[0064] In the refrigeration apparatus described in Patent Literature 1 and the like, a method of reducing power consumption when one type of refrigerant is circulated is proposed. However, it is difficult to directly apply the method described in Patent Literature 1 to a refrigeration cycle system that performs air conditioning or the like by a plurality of refrigeration cycles using different types of refrigerants. There is a problem of reducing power consumption in a refrigeration cycle system that includes a cascade heat exchanger that causes heat exchange between a CO.sub.2 refrigerant and a refrigerant other than CO.sub.2 and that implements a first refrigeration cycle using the CO.sub.2 refrigerant and a second refrigeration cycle using the refrigerant other than CO.sub.2.

[0065] A refrigeration cycle system according to a first aspect includes a first refrigerant circuit, a second refrigerant circuit, and a controller. The first refrigerant circuit includes a first compressor that circulates a first refrigerant at a discharge refrigerant pressure in a range of 0.5 MPa or more and 4 MPa or less, and a cascade heat exchanger that cools a second refrigerant that is a CO.sub.2 refrigerant by the first refrigerant, and implements a first vapor compression refrigeration cycle using the first refrigerant. The second refrigerant circuit includes a second compressor that

circulates the second refrigerant at a discharge refrigerant pressure in a range of 5 MPa or more and 14 MPa or less, and implements a second vapor compression refrigeration cycle using the second refrigerant. The controller controls the first compressor and the second compressor. The controller controls the first compressor and the second compressor such that an intermediate temperature between an evaporation temperature of the first refrigerant and a condensation temperature of the second refrigerant in the cascade heat exchanger satisfies $T_2 + (T_1 - T_2) \times 0.1 \leq T_V \leq T_2 + (T_1 - T_2) \times 0.4$, where T_1 ° C. represents a pressure equivalent saturation temperature of a refrigerant discharged from the first compressor, and T_2 ° C. represents a pressure equivalent saturation temperature of a refrigerant sucked into the second compressor.

[0066] In the refrigeration cycle system according to the first aspect, the first refrigerant circuit including the first compressor more efficient than the second compressor transports a larger amount of heat, and thus, power consumption as a whole can be reduced.

[0067] A refrigeration cycle system according to a second aspect is the system according to the first aspect, in which the first refrigerant is R32, R454C, propane, R1234yf, R1234ze, or ammonia, or a refrigerant including any of R32, R454C, propane, R1234yf, R1234ze, or ammonia.

[0068] A refrigeration cycle system according to a third aspect is the system according to the first or second aspect, in which the second refrigerant circuit condenses the second refrigerant in the cascade heat exchanger.

[0069] A refrigeration cycle system according to a fourth aspect is the system according to the third aspect, in which the second refrigerant circuit includes an indoor heat exchanger that causes heat exchange between indoor air and the second refrigerant.

[0070] A refrigeration cycle system according to a fifth aspect is the system according to the fourth aspect, in which the first refrigerant circuit includes an outdoor heat exchanger that causes heat exchange between outside air and the first refrigerant.

[0071] The embodiment of the present disclosure has been described above. It will be understood that various changes to modes and details can be made without departing from the gist and scope of the present disclosure recited in the claims.

REFERENCE SIGNS LIST

[0072] **1** refrigeration cycle system [0073] **100** first refrigerant circuit [0074] **110** first compressor [0075] **130** outdoor air heat exchanger (example of outdoor heat exchanger) [0076] **150** cascade heat exchanger [0077] **200** second refrigerant circuit [0078] **210** second compressor [0079] **251, 252, 253** indoor air heat exchanger (example of indoor heat exchanger) [0080] **300** controller

CITATION LIST

Patent Literature

[0081] Patent Literature 1: JP 2002-147819 A

Claims

1. A refrigeration cycle system comprising: a first refrigerant circuit including a first compressor that circulates a first refrigerant at a discharge refrigerant pressure in a range of 0.5 MPa or more and 4 MPa or less and a cascade heat exchanger that cools a second refrigerant that is a CO₂ refrigerant by the first refrigerant, the first refrigeration circuit implementing a first vapor compression refrigeration cycle using the first refrigerant; a second refrigerant circuit including a second compressor that circulates the second refrigerant at a discharge refrigerant pressure in a range of 5 MPa or more and 14 MPa or less, the second refrigerant circuit implementing a second vapor compression refrigeration cycle using the second refrigerant; and a controller that controls the first compressor and the second compressor, wherein the controller controls the first compressor and the second compressor such that an intermediate temperature T_V between an evaporation temperature of the first refrigerant and a condensation temperature of the second refrigerant in the cascade heat exchanger satisfies

- $T2+(T1-T2)\times 0.1\leq TV\leq T2+(T1-T2)\times 0.4$ in which a pressure equivalent saturation temperature in ° C. of a refrigerant discharged from the first compressor is T1 and a pressure equivalent saturation temperature in ° C. of a refrigerant sucked into the second compressor is T2.
2. The refrigeration cycle system according to claim 1, wherein the first refrigerant includes R32, R454C, propane, R1234yf, R1234ze, or ammonia, or a refrigerant including any of R32, R454C, propane, R1234yf, R1234ze, or ammonia.
 3. The refrigeration cycle system according to claim 1, wherein the second refrigerant circuit condenses the second refrigerant in the cascade heat exchanger.
 4. The refrigeration cycle system according to claim 3, wherein the second refrigerant circuit includes an indoor heat exchanger that causes heat exchange between indoor air and the second refrigerant.
 5. The refrigeration cycle system according to claim 4, wherein the first refrigerant circuit includes an outdoor heat exchanger that causes heat exchange between outside air and the first refrigerant.
 6. The refrigeration cycle system according to claim 2, wherein the second refrigerant circuit condenses the second refrigerant in the cascade heat exchanger.
 7. The refrigeration cycle system according to claim 6, wherein the second refrigerant circuit includes an indoor heat exchanger that causes heat exchange between indoor air and the second refrigerant.
 8. The refrigeration cycle system according to claim 7, wherein the first refrigerant circuit includes an outdoor heat exchanger that causes heat exchange between outside air and the first refrigerant.
 9. The refrigeration cycle system according to claim 1, wherein a total power consumption of the refrigeration cycle system is 9.5 kW or less.
 10. The refrigeration cycle system according to claim 1, wherein a total power consumption of the refrigeration cycle system is 9.4 kW or less.
 11. The refrigeration cycle system according to claim 1, further comprising: a high-pressure pressure sensor on a discharge side of the first compressor, the high-pressure pressure sensor detecting the pressure equivalent saturation temperature T1 of refrigerant discharged from the first compressor; and a low-pressure pressure sensor on a suction side of the second compressor, the low-pressure pressure sensor detecting the pressure equivalent saturation temperature T2 of refrigerant sucked into the second compressor.
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